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Unit – 1 : Session – 5 : SLO – 1 | Dynamic Memory Allocation

■ I. Fill in the Blanks

1. run — dynamic memory is allocated at run time.
2. allocate — malloc() allocates raw memory.
3. deallocate (release) — free() releases memory.
4. initializes — calloc() allocates and zero-initializes memory.
5. resize — realloc() resizes previously allocated memory.
6. the size (in bytes) — sizeof() gives the size of a data type.
7. freed — a memory leak happens if memory is not freed.
8. void* (generic) — malloc() returns a void pointer.
9. malloc() — type casting is needed when using malloc in C (void* → target type).
10. fails — NULL is returned if allocation fails.
11. 5 — malloc(5 * sizeof(int)) allocates memory for 5 integers.
12. dereference (*) / arrow — dynamic memory is accessed via the dereference or array operator.
13. heap — dynamic arrays use heap memory.
14. stack (static) — static arrays use stack/static memory.
15. use-after-free (dangling pointer access).

■ II. Code Snippets

1. `int *arr = (int*) malloc(10 * sizeof(int));`
2. `float *f = (float*) calloc(5, sizeof(float));`
3. `arr = (int*) realloc(arr, 20 * sizeof(int));`
4. `free(arr); arr = NULL;`
5. `if (arr == NULL) { printf("Allocation failed"); exit(1); }`